Report for Task 3 (2)

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PC2 Volatility Analysis

AR(1) Model for PC2 Volatility

Let V_t^{PC2} denote the annual PC2 volatility calculated from the log zero-coupon rates. Following the same procedure used for PC1, the natural logarithm of PC2 volatility was modeled as an AR(1) process:

$$\ln(V_t^{PC2}) = \alpha + \beta \ln(V_{t-1}^{PC2}) + Z_t.$$

The estimated model is

$$\boxed{\ln(V_t^{PC2}) = -0.65591 + 0.85026 \ln(V_{t-1}^{PC2}) + Z_t}$$

Coefficient	Estimate	Std. Error	p-value
α	-0.65591	0.29908	0.0321
β	0.85026	0.06615	< 0.001

Diagnostics of the Innovations

The innovations Z_t from the AR(1) model were examined for normality and serial dependence.

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.0008111	Reject normality
Jarque-Bera	0.0000125	Reject normality
Ljung-Box Z_t, lag 5	0.7932	Fail to reject
Ljung-Box Z_t, lag 10	0.9170	Fail to reject
Ljung-Box $\lvert Z_t \rvert$, lag 5	0.7982	Fail to reject
Ljung-Box $\lvert Z_t \rvert$, lag 10	0.5949	Fail to reject

The Ljung-Box tests provide no evidence of statistically significant serial dependence in either Z_t or $|Z_t|$. However, both the Shapiro-Wilk and Jarque-Bera tests strongly reject normality. Therefore, the original innovations exhibit behavior consistent with independence under these diagnostics, but they are not Gaussian.

Normalization by S&P Volatility

The innovations were normalized using annual S&P volatility.

Let V_t^{SP} denote S&P volatility. Dividing the original AR(1) equation by V_t^{SP} gives

$$\frac{\ln(V_t^{PC2})}{V_t^{SP}} = \frac{\alpha}{V_t^{SP}} + \beta \frac{\ln(V_{t-1}^{PC2})}{V_t^{SP}} + \frac{Z_t}{V_t^{SP}}.$$

Let the normalized innovation now be;

$$W_t = \frac{Z_t}{V_t^{SP}}.$$

Diagnostics After S&P Volatility Normalization

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.2248	Fail to reject normality
Jarque-Bera	0.1006	Fail to reject normality
Ljung-Box W_t, lag 5	0.4777	Fail to reject
Ljung-Box W_t, lag 10	0.5649	Fail to reject
Ljung-Box $\|W_t\|$, lag 5	0.5444	Fail to reject
Ljung-Box $\|W_t\|$, lag 10	0.3006	Fail to reject

After normalization by S&P volatility, both the Shapiro-Wilk and Jarque-Bera tests fail to reject normality at the 5% significance level. The Ljung-Box tests also fail to detect statistically significant serial dependence in either W_t or $|W_t|$.

Therefore, the normalized innovations are consistent with IID Gaussian behavior.

Conclusion

The original PC2 innovations display no significant serial dependence, but normality is strongly rejected. After dividing the innovations by annual S&P volatility, normality is no longer rejected and there is still no evidence of serial dependence.

Thus, normalization by stock-market volatility substantially improved the innovation properties and we can conlcude that:

$$\boxed{W_t = \frac{Z_t}{V_t^{SP}} \text{ is consistent with IID Gaussian behavior.}}$$